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# CMPU Aged Order Analysis

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## 1 Executive Summary

In CMPU (Customer Must Upload) vet diet orders shipped or canceled between May 1, 2026 and July 31, 2026, 38,314 orders were canceled due to aging. In this context, aging means the time between order creation and order shipment or cancellation exceeded 11 days. CMPU orders are particular in that the customers must upload the prescription documents themselves for approval.

The number of aging orders was about 10% of the orders that were actually shipped with an estimated loss of 3,5 million, again about 10% of the net sales of the actually shipped orders (Table 1).

In the following seven days after a cancellation, 2,316 out of 30,382 customers return to shop the same part number for the same pet and successfully reorder for a total net sales of 221 thousand (Table 2) recovering 7.6% of the loss. After 28 days, Chewy recovers only about 20% of the estimated net sales. By 70 days, we have only 26% of the customers shopping the same product for the same pet. The attrition is worse for customers whose first CMPU order was the aged order; retention is 7.2% and 21.72% respectively for 7 and 70 days (11,500 customers, see Table 3). The worst segment is the new to vet diet customers; and their 70 day reorder rate is only 15% (5K customers, see Table 4). The rate of reorder in Chewy of these cohorts follows the same order 61%, 58% and 50%. A subsequent study will analyze longer time frames.

It is alarming that new vet diet customers suffer attrition significantly more.

All of the customers seem to have contacted Chewy, most of them at the creation of the order (Table 15). Some of them have contacted Chewy on multiple days (Table 16). People who contacted closer to the aged order cancellation date have a higher reorder rate (Table 18).

The study analyzes first orders and order lines associated with CMPU prescriptions. Details can be found in Section 2.4. The main findings are that highly likely aged orders are caused by customers not uploading their prescription documents; and that after 3 days there is 50%+ chance that a CMPU order becomes an aged order.

That the attrition rate is higher for customers who encounter this first time hints that uploading the prescription is an onus on the customer. The customer might not be uploading the prescription because it might be hard to obtain from the vet in person, it might be hard to upload for some reason ( the Chewy web site or Chewy app not working for the customer), or once the customer goes to the vet, they have the ability procure the product on site or nearby.

One suggestion to increase the upload rate (communication with Courtney Gleede) is to send reminder emails to customer. At this point, Chewy does not contact customers if there is a chance that their order might become aged. There is no Chewy relevant outbound contact data for a comparative/causal analysis. The only contact happens when the customer is notified that their order has been canceled. Assuming that people who receive an email before cancellation, react similarly to the cohort that reorders within 7 days after cancellation, we expect that out of 30K people considered in the opportunity study with aged cancellations, 2.3K would react before order cancellation corresponding to 221K sales. The best effort guess is that Chewy prevents people from skipping one reorder for this cohort for an additional gain of 221K net sales. The upper bound on the best effort estimate is 390K corresponding to the 14 day cohort. This represents about 1% increase in net sales for CMPU vet diet orders.

The effect of reminder emails on long term retention is an open question. Uploading a prescription is still an onus with or without a reminder.

### 1.1 Suggested Next Steps

- Chewy should contact customers who had aged orders in person or via AI (AI calls or emails) to understand the reasons of why customers have not uploaded their prescriptions and any other difficulty they may have encountered in the process. Chewy should focus on first time vet diet customers, as their retention is the worst.
- A controlled experiment where Chewy sends reminder emails to get a better understanding of the entitlement regarding short term gain is suggested. The suggestion is to pick a cohort of new vet diet customers who need to upload their prescriptions, randomly split it in half and send an email to one half to observe if Chewy can reduce the aged orders in the treatment group (customers to whom an email is sent) compared to the control ( no email sent).

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## 2.1 Reproducing the Numerical Results

The code to reproduce all the results is in [1]. Follow the instructions in the HOWTO.md file to generate the figures, models and analyses. You will need to clone the repository to run the code. The code uses make and python.

The tables and grains are TBD.

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## 2.2 Financial Opportunity Analysis

Table 1 reports distinct customer and order counts, qualifying order-line counts and recorded net sales by cancellation reason for orders shipped or cancelled from May 1, 2026 through July 31, 2026, inclusive, UTC (`[2026-05-01, 2026-08-01)`, with an exclusive end date). The population requires vet-diet lines matched by prescription ID and numeric customer ID to vet-diet prescriptions with `PRESCRIPTION_SOURCE = CMPU` and `APPROVAL_CHANNEL = CUSTOMER_UPLOAD`.

### 2.2.1 Orders by cancellation reason

Table 1: Financial opportunity baseline: cancellation reasons, customers, orders, qualifying order lines and net sales for CMPU vet-diet customer-upload orders shipped or cancelled May 1, 2026 through July 31, 2026, inclusive, UTC.

| Cancellation reason | Customers | Orders | Order lines | Net sales |
|---|---|---|---|---|
| [No recorded cancellation reason] | 172,878 | 387,707 | 433,051 | 35,821,384.85 |
| Aged order | 30,382 | 38,314 | 41,713 | 3,470,380.19 |
| Customer initiated cancellation | 29,455 | 32,939 | 37,959 | 2,945,818.13 |
| Customer request | 11,382 | 11,948 | 13,662 | 1,042,195.19 |
| Duplicate order | 161 | 165 | 188 | 14,763.64 |
| Fraud | 36 | 38 | 45 | 4,339.97 |
| Inventory issue | 5 | 5 | 5 | 583.24 |
| No response | 1,035 | 1,229 | 1,329 | 111,509.37 |
| Other | 784 | 809 | 940 | 71,158.59 |
| Payment declined | 4,242 | 5,533 | 6,067 | 523,931.07 |
| Payment issue | 46 | 47 | 57 | 4,207.82 |
| Pharmacy rejection | 278 | 310 | 351 | 29,055.56 |
| Prescription issue | 5,051 | 5,329 | 5,822 | 480,281.17 |
| Restricted availability | 1 | 1 | 1 | 25.64 |
| TOTAL | 225,434 | 484,374 | 541,190 | 44,519,634.43 |

*Notes:* Includes all qualifying orders, including subsequent orders; shipment or cancellation records must fall in the reporting window. There is no order-placement or prescription-creation restriction. This differs from the prescription-creation cohort and aged-cancellation cutoff used in Table 19. Counts are distinct customers, orders and physical order lines. Customers may appear in multiple reason rows; TOTAL counts each customer once. Sales include only qualifying lines, summed from `ORDER_LINE_NET_SALES` in `EDLDB.BT_HCA_HCDM.ORDER_LINE_COST_MEASURES_RXP`, once per shipment-and-line record with a shipment or cancellation timestamp in the window. Split-shipment sales outside the window are excluded; records meeting both dates count once. Zero and negative sales remain included; any missing sales make the group total incomplete. Display rounds to two decimals; exports retain five. Current cancellation reasons are grouped case-insensitively; a missing reason does not establish an uncancelled order. Orders can appear in multiple reason rows if their lines have different reasons; TOTAL counts each order once. Line counts and sales are additive. Gray net-sales amounts have a recorded cancellation reason and represent unrealized sales, not realized revenue. TOTAL includes these gray amounts. Recorded net sales are a baseline for opportunity sizing, not an estimate of recoverable revenue from cancelled orders.

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### 2.2.2 Non-cancelled reorders after aged cancellation

Table 2 summarizes cumulative follow-up windows after CMPU vet-diet `CUSTOMER_UPLOAD` aged cancellations from May 1, 2026 through July 31, 2026, inclusive, UTC.

Table 2: Customers, cohort percentages, non-cancelled reorder orders and matching-line net sales within 7, 14, 21, 28, 35, 42, 49, 56, 63 and 70 days after aged cancellation.

| Follow-up (days) | Customers | Cohort (%) | Orders | Net sales |
|---|---|---|---|---|
| 7 | 2,316 | 7.62% | 2,407 | 221,122.35 |
| 14 | 3,832 | 12.61% | 4,143 | 386,124.68 |
| 21 | 5,133 | 16.89% | 5,951 | 558,914.70 |
| 28 | 5,850 | 19.25% | 7,289 | 685,719.76 |
| 35 | 6,461 | 21.27% | 8,905 | 841,298.69 |
| 42 | 6,818 | 22.44% | 10,118 | 958,671.19 |
| 49 | 7,280 | 23.96% | 11,898 | 1,131,240.46 |
| 56 | 7,510 | 24.72% | 13,152 | 1,252,558.38 |
| 63 | 7,756 | 25.53% | 14,621 | 1,393,281.96 |
| 70 | 7,944 | 26.15% | 15,808 | 1,507,063.94 |

*Notes:* Match exact customer ID, `PRODUCT_PART_NUMBER` and non-null `PET_ID` on a different order, placed strictly after cancellation through the indicated number of days, inclusive. Original eligibility is selected by cancellation date, with no prescription-creation or first-order restriction. All windows use the same source snapshot and require complete follow-up through 2026-10-28 00:00 UTC. Reorders may use any approval channel or prescription source within the certified RxP order-line view. Exclude orders if any line has status X, a recorded cancellation reason or cancellation timestamp; also exclude orders with unknown line status. Statuses reflect the extraction snapshot. Each customer and reorder order counts once within a window, even when linked to several original cancellations. Windows and studies overlap; do not add their counts. Net sales sum `ORDER_LINE_NET_SALES` on matching reorder lines only, once per shipment/line contribution, preserving zero and negative amounts. Sales may be recorded after the placement window; this is recorded matching-line sales, not proof of recovered revenue. Missing sales make the total incomplete. Every cohort percentage divides returning customers by the same full aged-customer cohort of 30,382, including customers with no qualifying reorder or unusable matching keys. A zero cohort is shown as N/A. Amounts display two decimals; exports retain five.

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### 2.2.3 Follow-up for first CUSTOMER_UPLOAD vet-diet customers

Table 3 follows the 11,550 customers whose first observed vet-diet CUSTOMER_UPLOAD order was an aged order. Original CMPU vet-diet CUSTOMER_UPLOAD aged cancellations occurred from May 1, 2026 through July 31, 2026, inclusive, UTC.

Table 3: Same-pet, same-product non-cancelled reorders after cancellation of a first CUSTOMER_UPLOAD vet-diet aged order: cumulative customers, percentage of the fixed cohort, orders and matching-line net sales through 70 days.

| Follow-up (days) | Customers | Cohort (%) | Orders | Net sales |
|---|---|---|---|---|
| 7 | 833 | 7.21% | 855 | 72,184.56 |
| 14 | 1,281 | 11.09% | 1,365 | 116,938.34 |
| 21 | 1,633 | 14.14% | 1,840 | 159,141.20 |
| 28 | 1,846 | 15.98% | 2,256 | 193,780.75 |
| 35 | 2,026 | 17.54% | 2,698 | 234,112.61 |
| 42 | 2,149 | 18.61% | 3,113 | 272,143.01 |
| 49 | 2,297 | 19.89% | 3,641 | 322,633.43 |
| 56 | 2,372 | 20.54% | 3,995 | 354,230.43 |
| 63 | 2,450 | 21.21% | 4,425 | 393,797.77 |
| 70 | 2,509 | 21.72% | 4,801 | 427,522.19 |

*Notes:* Each customer's first qualifying aged order is the sole anchor; later aged orders do not reset follow-up. Follow-up begins strictly after that order's cancellation and includes the indicated day limit. Require the same numeric customer ID, exact PRODUCT_PART_NUMBER and non-null PET_ID. Later orders may use any approval channel or prescription source. Exclude the whole later order if any line is cancelled or has unknown status. Count each customer, order and physical line once within each window, regardless of multiple original lines or shipments. Every percentage uses the same fixed denominator of 11,550 customers, including those with no qualifying reorder. All windows require complete follow-up through 2026-10-28 00:00 UTC; windows and studies overlap and should not be added. The cohort uses exact earliest observed order IDs in linked prescription and ECOM history, across all historical prescription sources, with current prescription approval-channel snapshots. Original eligibility is CMPU, vet diet and CUSTOMER_UPLOAD. Current extraction statuses determine non-cancellation. Net sales sum ORDER_LINE_NET_SALES on matching lines, once per shipment/line contribution; missing sales make totals incomplete. Sales are recorded matching-line amounts, not proof of recovered revenue. Amounts display two decimals; exports retain five.

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### 2.2.4 Follow-up for first vet-diet customers

Table 4 follows the 5,097 customers whose first observed vet-diet order was an aged order.

Original CMPU vet-diet `CUSTOMER_UPLOAD` aged cancellations occurred from May 1, 2026 through July 31, 2026, inclusive, UTC.

Table 4: Same-pet, same-product non-cancelled reorders after cancellation of a first vet-diet aged order: cumulative customers, percentage of the fixed cohort, orders and matching-line net sales through 70 days.

| Follow-up (days) | Customers | Cohort (%) | Orders | Net sales |
|---|---|---|---|---|
| 7 | 312 | 6.12% | 322 | 23,939.00 |
| 14 | 468 | 9.18% | 500 | 36,606.75 |
| 21 | 565 | 11.08% | 642 | 47,225.57 |
| 28 | 631 | 12.38% | 792 | 58,032.36 |
| 35 | 680 | 13.34% | 914 | 67,607.06 |
| 42 | 720 | 14.13% | 1,061 | 79,250.13 |
| 49 | 749 | 14.69% | 1,191 | 89,793.43 |
| 56 | 766 | 15.03% | 1,308 | 99,811.18 |
| 63 | 781 | 15.32% | 1,425 | 109,761.26 |
| 70 | 800 | 15.70% | 1,567 | 120,434.49 |

*Notes:* Each customer's first qualifying aged order is the sole anchor; later aged orders do not reset follow-up. Follow-up begins strictly after that order's cancellation and includes the indicated day limit. Require the same numeric customer ID, exact `PRODUCT_PART_NUMBER` and non-null `PET_ID`. Later orders may use any approval channel or prescription source. Exclude the whole later order if any line is cancelled or has unknown status. Count each customer, order and physical line once within each window, regardless of multiple original lines or shipments. Every percentage uses the same fixed denominator of 5,097 customers, including those with no qualifying reorder. All windows require complete follow-up through 2026-10-28 00:00 UTC; windows and studies overlap and should not be added. The cohort uses earliest observed vet-diet order IDs across all historical approval channels and prescription sources, identified from RxP vet-diet lines and ECOM order history. Original eligibility is CMPU, vet diet and `CUSTOMER_UPLOAD`. Current extraction statuses determine non-cancellation. Net sales sum `ORDER_LINE_NET_SALES` on matching lines, once per shipment/line contribution; missing sales make totals incomplete. Sales are recorded matching-line amounts, not proof of recovered revenue. Amounts display two decimals; exports retain five.

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### 2.2.5 Non-cancelled vet-diet reorders: same pet, different product

Table 5 summarizes cumulative follow-up windows after CMPU vet-diet `CUSTOMER_UPLOAD` aged cancellations from May 1, 2026 through July 31, 2026, inclusive, UTC.

Table 5: Same-pet, different-product vet-diet reorders: customers, cohort percentages, orders and net sales within 7, 14, 21, 28, 35, 42, 49, 56, 63 and 70 days after aged cancellation.

| Follow-up (days) | Customers | Cohort (%) | Orders | Net sales |
|---|---|---|---|---|
| 7 | 1,332 | 4.38% | 1,457 | 132,081.20 |
| 14 | 2,182 | 7.18% | 2,553 | 230,877.67 |
| 21 | 2,890 | 9.51% | 3,677 | 331,564.87 |
| 28 | 3,413 | 11.23% | 4,690 | 423,982.81 |
| 35 | 3,814 | 12.55% | 5,711 | 517,496.22 |
| 42 | 4,138 | 13.62% | 6,656 | 602,901.60 |
| 49 | 4,421 | 14.55% | 7,662 | 698,059.40 |
| 56 | 4,651 | 15.31% | 8,624 | 784,739.53 |
| 63 | 4,856 | 15.98% | 9,594 | 871,819.44 |
| 70 | 5,010 | 16.49% | 10,552 | 959,049.10 |

*Notes:* Use a different order placed strictly after cancellation through the indicated day limit, inclusive. Match exact customer ID and non-null `PET_ID`; require vet-diet lines with a nonblank `PRODUCT_PART_NUMBER` different from the original. Original eligibility is selected by cancellation date, with no prescription-creation or first-order restriction. All windows use the same source snapshot and require complete follow-up through 2026-10-28 00:00 UTC. Reorders may use any approval channel or prescription source within the certified RxP order-line view. Exclude orders if any line has status X, a recorded cancellation reason or cancellation timestamp; also exclude orders with unknown line status. Statuses reflect the extraction snapshot. Each customer and reorder order counts once within a window, even when linked to several original cancellations. Windows and studies overlap; do not add their counts. Net sales sum `ORDER_LINE_NET_SALES` on matching reorder lines only, once per shipment/line contribution, preserving zero and negative amounts. Sales may be recorded after the placement window; this is recorded matching-line sales, not proof of recovered revenue. Missing sales make the total incomplete. Every cohort percentage divides returning customers by the same full aged-customer cohort of 30,382, including customers with no qualifying reorder or unusable matching keys. A zero cohort is shown as N/A. Amounts display two decimals; exports retain five.

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### 2.2.6 Same-pet, different-product vet-diet follow-up for first upload customers

Table 6 follows the 11,550 customers whose first observed vet-diet CUSTOMER_UPLOAD order was an aged order. Original CMPU vet-diet CUSTOMER_UPLOAD aged cancellations occurred from May 1, 2026 through July 31, 2026, inclusive, UTC.

Table 6: Same-pet, different-product vet-diet non-cancelled reorders after cancellation of a first CUSTOMER_UPLOAD vet-diet aged order: cumulative customers, percentage of the fixed cohort, orders and matching-line net sales through 70 days.

| Follow-up (days) | Customers | Cohort (%) | Orders | Net sales |
|---|---|---|---|---|
| 7 | 444 | 3.84% | 466 | 40,916.08 |
| 14 | 727 | 6.29% | 792 | 68,360.26 |
| 21 | 924 | 8.00% | 1,089 | 94,359.87 |
| 28 | 1,080 | 9.35% | 1,379 | 119,958.61 |
| 35 | 1,217 | 10.54% | 1,695 | 147,193.63 |
| 42 | 1,314 | 11.38% | 1,983 | 173,509.35 |
| 49 | 1,401 | 12.13% | 2,289 | 201,286.86 |
| 56 | 1,479 | 12.81% | 2,574 | 225,931.09 |
| 63 | 1,553 | 13.45% | 2,893 | 253,413.39 |
| 70 | 1,617 | 14.00% | 3,220 | 281,008.90 |

*Notes:* Each customer's first qualifying aged order is the sole anchor; later aged orders do not reset follow-up. Follow-up begins strictly after that order's cancellation and includes the indicated day limit. Require the same numeric customer ID and non-null PET_ID, a different nonblank PRODUCT_PART_NUMBER from the anchor line, and a TRUE vet-diet flag on the later line. Later orders may use any approval channel or prescription source. Exclude the whole later order if any line is cancelled or has unknown status. Count each customer, order and physical line once within each window, regardless of multiple original lines or shipments. Every percentage uses the same fixed denominator of 11,550 customers, including those with no qualifying reorder. All windows require complete follow-up through 2026-10-28 00:00 UTC; windows and studies overlap and should not be added. The cohort uses exact earliest observed order IDs in linked prescription and ECOM history, across all historical prescription sources, with current prescription approval-channel snapshots. Original eligibility is CMPU, vet diet and CUSTOMER_UPLOAD. Current extraction statuses determine non-cancellation. Net sales sum ORDER_LINE_NET_SALES on matching lines, once per shipment/line contribution; missing sales make totals incomplete. Sales are recorded matching-line amounts, not proof of recovered revenue. Amounts display two decimals; exports retain five.

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### 2.2.7 Same-pet, different-product vet-diet follow-up for first vet-diet customers

Table 7 follows the 5,097 customers whose first observed vet-diet order was an aged order.

Original CMPU vet-diet `CUSTOMER_UPLOAD` aged cancellations occurred from May 1, 2026 through July 31, 2026, inclusive, UTC.

Table 7: Same-pet, different-product vet-diet non-cancelled reorders after cancellation of a first vet-diet aged order: cumulative customers, percentage of the fixed cohort, orders and matching-line net sales through 70 days.

| Follow-up (days) | Customers | Cohort (%) | Orders | Net sales |
|---|---|---|---|---|
| 7 | 181 | 3.55% | 189 | 15,801.67 |
| 14 | 276 | 5.41% | 307 | 24,589.86 |
| 21 | 332 | 6.51% | 399 | 32,725.31 |
| 28 | 382 | 7.49% | 505 | 40,896.95 |
| 35 | 423 | 8.30% | 605 | 48,954.22 |
| 42 | 451 | 8.85% | 704 | 57,525.03 |
| 49 | 479 | 9.40% | 800 | 65,677.94 |
| 56 | 504 | 9.89% | 896 | 73,577.63 |
| 63 | 530 | 10.40% | 1,010 | 83,296.37 |
| 70 | 548 | 10.75% | 1,122 | 91,928.98 |

*Notes:* Each customer's first qualifying aged order is the sole anchor; later aged orders do not reset follow-up. Follow-up begins strictly after that order's cancellation and includes the indicated day limit. Require the same numeric customer ID and non-null `PET_ID`, a different nonblank `PRODUCT_PART_NUMBER` from the anchor line, and a TRUE vet-diet flag on the later line. Later orders may use any approval channel or prescription source. Exclude the whole later order if any line is cancelled or has unknown status. Count each customer, order and physical line once within each window, regardless of multiple original lines or shipments. Every percentage uses the same fixed denominator of 5,097 customers, including those with no qualifying reorder. All windows require complete follow-up through 2026-10-28 00:00 UTC; windows and studies overlap and should not be added. The cohort uses earliest observed vet-diet order IDs across all historical approval channels and prescription sources, identified from RxP vet-diet lines and ECOM order history. Original eligibility is CMPU, vet diet and `CUSTOMER_UPLOAD`. Current extraction statuses determine non-cancellation. Net sales sum `ORDER_LINE_NET_SALES` on matching lines, once per shipment/line contribution; missing sales make totals incomplete. Sales are recorded matching-line amounts, not proof of recovered revenue. Amounts display two decimals; exports retain five.

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### 2.2.8 Non-cancelled vet-diet reorders: any pet or product

Table 8 summarizes cumulative follow-up windows after CMPU vet-diet `CUSTOMER_UPLOAD` aged cancellations from May 1, 2026 through July 31, 2026, inclusive, UTC.

Table 8: Any vet-diet reorders: customers, cohort percentages, orders and net sales within 7, 14, 21, 28, 35, 42, 49, 56, 63 and 70 days after aged cancellation.

| Follow-up (days) | Customers | Cohort (%) | Orders | Net sales |
|---|---|---|---|---|
| 7 | 3,594 | 11.83% | 4,020 | 376,580.78 |
| 14 | 5,824 | 19.17% | 7,052 | 666,449.99 |
| 21 | 7,637 | 25.14% | 10,163 | 960,248.17 |
| 28 | 8,713 | 28.68% | 12,680 | 1,202,863.81 |
| 35 | 9,555 | 31.45% | 15,448 | 1,470,197.68 |
| 42 | 10,075 | 33.16% | 17,772 | 1,694,190.10 |
| 49 | 10,631 | 34.99% | 20,632 | 1,969,933.29 |
| 56 | 10,976 | 36.13% | 23,017 | 2,198,639.92 |
| 63 | 11,314 | 37.24% | 25,580 | 2,445,610.75 |
| 70 | 11,567 | 38.07% | 27,878 | 2,665,884.12 |

*Notes:* Use a different order placed strictly after cancellation through the indicated day limit, inclusive. Match exact customer ID and require vet-diet lines; purchases may be for any pet and any product part number. Original eligibility is selected by cancellation date, with no prescription-creation or first-order restriction. All windows use the same source snapshot and require complete follow-up through 2026-10-28 00:00 UTC. Reorders may use any approval channel or prescription source within the certified RxP order-line view. Exclude orders if any line has status X, a recorded cancellation reason or cancellation timestamp; also exclude orders with unknown line status. Statuses reflect the extraction snapshot. Each customer and reorder order counts once within a window, even when linked to several original cancellations. Windows and studies overlap; do not add their counts. Net sales sum `ORDER_LINE_NET_SALES` on matching reorder lines only, once per shipment/line contribution, preserving zero and negative amounts. Sales may be recorded after the placement window; this is recorded matching-line sales, not proof of recovered revenue. Missing sales make the total incomplete. Every cohort percentage divides returning customers by the same full aged-customer cohort of 30,382, including customers with no qualifying reorder or unusable matching keys. A zero cohort is shown as N/A. Amounts display two decimals; exports retain five.

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### 2.2.9 Any vet-diet follow-up for first upload customers

Table 9 follows the 11,550 customers whose first observed vet-diet CUSTOMER_UPLOAD order was an aged order. Original CMPU vet-diet CUSTOMER_UPLOAD aged cancellations occurred from May 1, 2026 through July 31, 2026, inclusive, UTC.

Table 9: Any vet-diet non-cancelled reorders after cancellation of a first CUSTOMER_UPLOAD vet-diet aged order: cumulative customers, percentage of the fixed cohort, orders and matching-line net sales through 70 days.

| Follow-up (days) | Customers | Cohort (%) | Orders | Net sales |
|---|---|---|---|---|
| 7 | 1,217 | 10.54% | 1,305 | 115,132.09 |
| 14 | 1,914 | 16.57% | 2,178 | 192,777.87 |
| 21 | 2,432 | 21.06% | 3,005 | 267,367.22 |
| 28 | 2,766 | 23.95% | 3,743 | 332,421.02 |
| 35 | 3,051 | 26.42% | 4,540 | 405,245.66 |
| 42 | 3,240 | 28.05% | 5,282 | 474,927.09 |
| 49 | 3,423 | 29.64% | 6,127 | 555,536.18 |
| 56 | 3,547 | 30.71% | 6,825 | 618,447.80 |
| 63 | 3,664 | 31.72% | 7,624 | 693,176.04 |
| 70 | 3,753 | 32.49% | 8,395 | 762,808.17 |

*Notes:* Each customer's first qualifying aged order is the sole anchor; later aged orders do not reset follow-up. Follow-up begins strictly after that order's cancellation and includes the indicated day limit. Require the same numeric customer ID and a TRUE vet-diet flag on the later line, for any pet and product part number. Later orders may use any approval channel or prescription source. Exclude the whole later order if any line is cancelled or has unknown status. Count each customer, order and physical line once within each window, regardless of multiple original lines or shipments. Every percentage uses the same fixed denominator of 11,550 customers, including those with no qualifying reorder. All windows require complete follow-up through 2026-10-28 00:00 UTC; windows and studies overlap and should not be added. The cohort uses exact earliest observed order IDs in linked prescription and ECOM history, across all historical prescription sources, with current prescription approval-channel snapshots. Original eligibility is CMPU, vet diet and CUSTOMER_UPLOAD. Current extraction statuses determine non-cancellation. Net sales sum ORDER_LINE_NET_SALES on matching lines, once per shipment/line contribution; missing sales make totals incomplete. Sales are recorded matching-line amounts, not proof of recovered revenue. Amounts display two decimals; exports retain five.

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### 2.2.10 Any vet-diet follow-up for first vet-diet customers

Table 10 follows the 5,097 customers whose first observed vet-diet order was an aged order.

Original CMPU vet-diet `CUSTOMER_UPLOAD` aged cancellations occurred from May 1, 2026 through July 31, 2026, inclusive, UTC.

Table 10: Any vet-diet non-cancelled reorders after cancellation of a first vet-diet aged order: cumulative customers, percentage of the fixed cohort, orders and matching-line net sales through 70 days.

| Follow-up (days) | Customers | Cohort (%) | Orders | Net sales |
|---|---|---|---|---|
| 7 | 438 | 8.59% | 463 | 35,772.45 |
| 14 | 659 | 12.93% | 737 | 55,857.40 |
| 21 | 800 | 15.70% | 969 | 74,330.46 |
| 28 | 899 | 17.64% | 1,208 | 92,469.85 |
| 35 | 980 | 19.23% | 1,425 | 109,763.84 |
| 42 | 1,031 | 20.23% | 1,653 | 128,552.70 |
| 49 | 1,072 | 21.03% | 1,873 | 146,660.13 |
| 56 | 1,105 | 21.68% | 2,077 | 163,846.89 |
| 63 | 1,137 | 22.31% | 2,302 | 183,559.64 |
| 70 | 1,165 | 22.86% | 2,541 | 202,269.09 |

*Notes:* Each customer's first qualifying aged order is the sole anchor; later aged orders do not reset follow-up. Follow-up begins strictly after that order's cancellation and includes the indicated day limit. Require the same numeric customer ID and a TRUE vet-diet flag on the later line, for any pet and product part number. Later orders may use any approval channel or prescription source. Exclude the whole later order if any line is cancelled or has unknown status. Count each customer, order and physical line once within each window, regardless of multiple original lines or shipments. Every percentage uses the same fixed denominator of 5,097 customers, including those with no qualifying reorder. All windows require complete follow-up through 2026-10-28 00:00 UTC; windows and studies overlap and should not be added. The cohort uses earliest observed vet-diet order IDs across all historical approval channels and prescription sources, identified from RxP vet-diet lines and ECOM order history. Original eligibility is CMPU, vet diet and `CUSTOMER_UPLOAD`. Current extraction statuses determine non-cancellation. Net sales sum `ORDER_LINE_NET_SALES` on matching lines, once per shipment/line contribution; missing sales make totals incomplete. Sales are recorded matching-line amounts, not proof of recovered revenue. Amounts display two decimals; exports retain five.

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### 2.2.11 Non-cancelled follow-up purchases: all Chewy

Table 11 summarizes cumulative follow-up windows after CMPU vet-diet `CUSTOMER_UPLOAD` aged cancellations from May 1, 2026 through July 31, 2026, inclusive, UTC.

Table 11: All-Chewy follow-up purchases: customers, cohort percentages, non-cancelled orders and net sales within 7, 14, 21, 28, 35, 42, 49, 56, 63 and 70 days after aged cancellation.

| Follow-up (days) | Customers | Cohort (%) | Orders | Net sales |
|---|---|---|---|---|
| 7 | 6,477 | 21.32% | 8,512 | 702,794.12 |
| 14 | 10,265 | 33.79% | 15,918 | 1,307,265.44 |
| 21 | 13,193 | 43.42% | 23,645 | 1,912,710.30 |
| 28 | 14,804 | 48.73% | 30,064 | 2,440,684.77 |
| 35 | 15,917 | 52.39% | 36,535 | 2,971,072.85 |
| 42 | 16,689 | 54.93% | 42,616 | 3,471,768.55 |
| 49 | 17,370 | 57.17% | 49,096 | 4,009,870.68 |
| 56 | 17,837 | 58.71% | 55,126 | 4,500,734.83 |
| 63 | 18,249 | 60.07% | 61,158 | 4,996,504.86 |
| 70 | 18,569 | 61.12% | 66,946 | 5,473,570.84 |

*Notes:* Use a different order placed strictly after cancellation through the indicated day limit, inclusive. Match customer ID only; include all Chewy product categories and all pets. Original eligibility is selected by cancellation date, with no prescription-creation or first-order restriction. All windows use the same source snapshot and require complete follow-up through 2026-10-28 00:00 UTC. Follow-up orders come from `EDLDB.ECOM.ORDER_LINE`; no prescription, vet-diet, product or pet restriction applies. Exclude orders if any line has status X, a recorded cancellation reason or cancellation timestamp; also exclude orders with unknown line status. Statuses reflect the extraction snapshot. Each customer and reorder order counts once within a window, even when linked to several original cancellations. Windows and studies overlap; do not add their counts. Net sales sum `NET_SALES` on all qualifying ECOM order lines, once per physical order line, preserving zero and negative amounts. Exact source duplicates collapse and conflicting line values fail validation. Sales may be recorded after the placement window; this is recorded matching-line sales, not proof of recovered revenue. Missing sales make the total incomplete. Every cohort percentage divides returning customers by the same full aged-customer cohort of 30,382, including customers with no qualifying reorder or unusable matching keys. A zero cohort is shown as N/A. Amounts display two decimals; ECOM supplies four-decimal amounts, retained exactly in five-decimal exports.

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### 2.2.12 All-Chewy follow-up for first upload customers

Table 12 follows the 11,550 customers whose first observed vet-diet CUSTOMER_UPLOAD order was an aged order. Original CMPU vet-diet CUSTOMER_UPLOAD aged cancellations occurred from May 1, 2026 through July 31, 2026, inclusive, UTC.

Table 12: All-Chewy non-cancelled reorders after cancellation of a first CUSTOMER_UPLOAD vet-diet aged order: cumulative customers, percentage of the fixed cohort, orders and matching-line net sales through 70 days.

| Follow-up (days) | Customers | Cohort (%) | Orders | Net sales |
|---|---|---|---|---|
| 7 | 2,198 | 19.03% | 2,653 | 213,634.05 |
| 14 | 3,492 | 30.23% | 4,913 | 391,233.04 |
| 21 | 4,589 | 39.73% | 7,278 | 560,461.37 |
| 28 | 5,197 | 45.00% | 9,308 | 718,425.79 |
| 35 | 5,610 | 48.57% | 11,423 | 885,226.79 |
| 42 | 5,938 | 51.41% | 13,515 | 1,054,996.63 |
| 49 | 6,201 | 53.69% | 15,674 | 1,231,716.11 |
| 56 | 6,411 | 55.51% | 17,688 | 1,392,497.95 |
| 63 | 6,575 | 56.93% | 19,779 | 1,560,537.97 |
| 70 | 6,712 | 58.11% | 21,917 | 1,732,549.06 |

*Notes:* Each customer's first qualifying aged order is the sole anchor; later aged orders do not reset follow-up. Follow-up begins strictly after that order's cancellation and includes the indicated day limit. Require the same numeric customer ID; include all Chewy product categories and any pet/product, with no later prescription, approval-channel or source restriction. Later orders may use any approval channel or prescription source. Exclude the whole later order if any line is cancelled or has unknown status. Count each customer, order and physical line once within each window, regardless of multiple original lines or shipments. Every percentage uses the same fixed denominator of 11,550 customers, including those with no qualifying reorder. All windows require complete follow-up through 2026-10-28 00:00 UTC; windows and studies overlap and should not be added. The cohort uses exact earliest observed order IDs in linked prescription and ECOM history, across all historical prescription sources, with current prescription approval-channel snapshots. Original eligibility is CMPU, vet diet and CUSTOMER_UPLOAD. Current extraction statuses determine non-cancellation. Net sales sum NET_SALES from EDLDB.ECOM.ORDER_LINE on matching lines, once per physical order line; missing sales make totals incomplete. Sales are recorded matching-line amounts, not proof of recovered revenue. Amounts display two decimals; ECOM supplies four decimals, retained exactly in five-decimal exports.

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### 2.2.13 All-Chewy follow-up for first vet-diet customers

Table 13 follows the 5,097 customers whose first observed vet-diet order was an aged order.

Original CMPU vet-diet `CUSTOMER_UPLOAD` aged cancellations occurred from May 1, 2026 through July 31, 2026, inclusive, UTC.

Table 13: All-Chewy non-cancelled reorders after cancellation of a first vet-diet aged order: cumulative customers, percentage of the fixed cohort, orders and matching-line net sales through 70 days.

| Follow-up (days) | Customers | Cohort (%) | Orders | Net sales |
|---|---|---|---|---|
| 7 | 826 | 16.21% | 971 | 74,076.57 |
| 14 | 1,310 | 25.70% | 1,784 | 130,104.51 |
| 21 | 1,780 | 34.92% | 2,685 | 179,943.04 |
| 28 | 2,035 | 39.93% | 3,448 | 232,948.35 |
| 35 | 2,183 | 42.83% | 4,142 | 282,824.63 |
| 42 | 2,306 | 45.24% | 4,921 | 340,115.01 |
| 49 | 2,402 | 47.13% | 5,643 | 393,973.18 |
| 56 | 2,482 | 48.70% | 6,370 | 449,207.78 |
| 63 | 2,539 | 49.81% | 7,065 | 501,838.82 |
| 70 | 2,591 | 50.83% | 7,865 | 561,647.94 |

*Notes:* Each customer's first qualifying aged order is the sole anchor; later aged orders do not reset follow-up. Follow-up begins strictly after that order's cancellation and includes the indicated day limit. Require the same numeric customer ID; include all Chewy product categories and any pet/product, with no later prescription, approval-channel or source restriction. Later orders may use any approval channel or prescription source. Exclude the whole later order if any line is cancelled or has unknown status. Count each customer, order and physical line once within each window, regardless of multiple original lines or shipments. Every percentage uses the same fixed denominator of 5,097 customers, including those with no qualifying reorder. All windows require complete follow-up through 2026-10-28 00:00 UTC; windows and studies overlap and should not be added. The cohort uses earliest observed vet-diet order IDs across all historical approval channels and prescription sources, identified from RxP vet-diet lines and ECOM order history. Original eligibility is CMPU, vet diet and `CUSTOMER_UPLOAD`. Current extraction statuses determine non-cancellation. Net sales sum `NET_SALES` from `EDLDB.ECOM.ORDER_LINE` on matching lines, once per physical order line; missing sales make totals incomplete. Sales are recorded matching-line amounts, not proof of recovered revenue. Amounts display two decimals; ECOM supplies four decimals, retained exactly in five-decimal exports.

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### 2.2.14 Same-pet, same-product orders per returning customer

Table 14 shows the percentage distribution of distinct orders per returning customer for the same-pet, same-product follow-up in Table 2. The original CMPU vet-diet `CUSTOMER_UPLOAD` aged cancellations occurred from May 1, 2026 through July 31, 2026, inclusive, UTC.

Table 14: Same-pet, same-product follow-up: percentage of final 70-day returning customers by number of distinct non-cancelled orders.

| Follow-up (days) | Number of orders per customer | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| 7 | 28.03% | 1.10% | 0.03% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |
| 14 | 44.60% | 3.37% | 0.25% | 0.01% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |
| 21 | 55.63% | 7.85% | 0.97% | 0.15% | 0.01% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |
| 28 | 58.74% | 12.39% | 1.94% | 0.47% | 0.11% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |
| 35 | 57.74% | 18.09% | 4.29% | 0.82% | 0.31% | 0.08% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |
| 42 | 54.83% | 23.31% | 5.82% | 1.18% | 0.43% | 0.19% | 0.05% | 0.01% | 0.00% | 0.00% | 0.00% | 0.00% |
| 49 | 50.23% | 30.38% | 7.20% | 2.68% | 0.74% | 0.20% | 0.18% | 0.03% | 0.01% | 0.00% | 0.00% | 0.00% |
| 56 | 47.28% | 31.72% | 10.45% | 3.26% | 1.07% | 0.38% | 0.23% | 0.13% | 0.01% | 0.01% | 0.00% | 0.00% |
| 63 | 43.61% | 34.05% | 13.03% | 3.69% | 2.09% | 0.63% | 0.21% | 0.20% | 0.10% | 0.01% | 0.01% | 0.00% |
| 70 | 41.10% | 34.59% | 15.70% | 4.66% | 2.34% | 0.81% | 0.37% | 0.21% | 0.13% | 0.08% | 0.01% | 0.01% |

*Notes:* For each follow-up window, divide the customers with the indicated number of distinct qualifying orders by the same final 70-day total of 7,944 returning customers in Table 2, then multiply by 100. No zero-order column is shown. Match exact customer, `PET_ID` and `PRODUCT_PART_NUMBER`; use the same non-cancelled-order rules and complete follow-up through 2026-10-28 00:00 UTC. Count each order once per customer, regardless of lines, shipments or repeated original cancellations. Pool all qualifying matching pets/products and cancellation windows per customer. Earlier rows sum to the percentage of final-window customers who have reordered by that day; only the final row totals 100% before rounding. N/A denotes a zero final-window customer count. Windows overlap.

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## 2.3 Contact Analysis

### 2.3.1 Contacts by day relative to cancellation

Table 15 summarizes inbound contacts linked to CMPU vet-diet `CUSTOMER_UPLOAD` aged orders cancelled from May 1, 2026 through July 31, 2026, inclusive, UTC. The full cohort contains 30,382 customers and 38,314 aged orders.

Table 15: Customer-initiated inbound contacts by day relative to aged-order cancellation: distinct customers contacting and distinct aged order IDs, from 13 days before through day 3 after cancellation. A total of 30,299 distinct customers contacted across this window.

| Days relative to cancellation | Customers contacting | Aged order IDs |
|---:|---:|---:|
| -13 | 38 | 38 |
| -12 | 858 | 861 |
| -11 | 19,267 | 23,158 |
| -10 | 11,364 | 13,530 |
| -9 | 571 | 575 |
| -8 | 215 | 215 |
| -7 | 143 | 143 |
| -6 | 123 | 123 |
| -5 | 106 | 106 |
| -4 | 99 | 99 |
| -3 | 128 | 128 |
| -2 | 251 | 256 |
| -1 | 248 | 248 |
| 0 | 71 | 71 |
| 1 | 53 | 53 |
| 2 | 28 | 28 |
| 3 | 22 | 23 |

*Notes:* Customer-initiated is defined by `SESSION_INITIAL_DIRECTION=inbound` in `EDLDB.BT_CUSTOMER_CARE.FCT_CONTACTS_BRD`, across all contact categories. Include handled, unhandled and unknown-handled inbound attempts. Match the original aged `ORDER_ID` and require the contact's numeric `CUSTOMER_ID` to match the order's customer. Use UTC `SESSION_START_DTTM`. Day $n$ is the elapsed 24-hour interval $[n, n+1)$ days relative to cancellation: day $-1$ is the preceding 24 hours, and day $0$ starts exactly at cancellation. These are elapsed bins rather than calendar-date differences. All seventeen rows from $-13$ through $+3$ are retained, with zero for missing days. Count each customer and aged order once per day, regardless of repeat contact legs, sessions or qualifying order lines. Counts may overlap across days and should not be added as global unique totals. Missing timestamps or customer IDs, mismatching customer IDs, unknown directions and outbound contacts remain audited and are excluded from this table. Email contacts with Other ingress are included; inbound direction is a source proxy and does not establish human initiation. Full channel/ingress audits are exported with the data. Original eligibility uses cancellation dates and all qualifying orders, without a first-order or prescription-creation restriction. Observation cutoff: 2026-10-28 00:00 UTC, allowing the full positive-day window.

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### 2.3.2 Customer contact-day frequency

Table 16 groups customers by the number of distinct relative days on which they contacted during the same $-13$ through $+3$ window.

Table 16: Frequency of customer contact days around aged-order cancellation: number of distinct relative days contacted versus number of customers. Of 30,382 cohort customers, 30,299 contacted on at least one day.

| Number of days contacted | Number of customers |
|---:|---:|
| 0 | 83 |
| 1 | 27,265 |
| 2 | 2,808 |
| 3 | 201 |
| 4 | 24 |
| 5 | 1 |
| 6 | 0 |
| 7 | 0 |
| 8 | 0 |
| 9 | 0 |
| 10 | 0 |
| 11 | 0 |
| 12 | 0 |
| 13 | 0 |
| 14 | 0 |
| 15 | 0 |
| 16 | 0 |
| 17 | 0 |

*Notes:* Same CMPU vet-diet `CUSTOMER_UPLOAD` aged cohort, cancellation dates, observation cutoff and inbound contact rules as Table 15. Count each distinct relative 24-hour day bin from $-13$ through $+3$ once per customer across all qualifying aged orders. Multiple contacts, sessions, legs or orders on the same relative day do not increase the day count. These are relative cancellation-day bins, rather than distinct calendar dates. Each customer appears in exactly one frequency row; zero includes cohort customers with no qualifying contact in this window. All frequencies from 0 through 17 are retained, including zero-count rows. Customer counts sum to the full cohort; positive-day rows sum to the distinct contacting customers.

### 2.3.3 Reorders by contact-day frequency

Table 17 applies the contact-day frequency segments in Table 16 to same-pet, same-product reorders.

Table 17: Same-pet, same-product-part-number reorders within 14 days after aged cancellation, by customer contact-day frequency. Original CMPU vet-diet `CUSTOMER_UPLOAD` orders were cancelled from May 1, 2026 through July 31, 2026, inclusive, UTC.

| Days contacted | Segment customers | Customers reordered | Segment reordered (%) |
|---:|---:|---:|---:|
| 1 | 27,265 | 3,376 | 12.38% |
| 2 | 2,808 | 396 | 14.10% |

*Notes:* Segments count distinct relative 24-hour contact-day bins from $-13$ through $+3$, pooled across each customer's qualifying aged orders. Multiple contacts on the same relative day count as one day. Match inbound contacts to the original aged order and its customer, across all categories. Include handled, unhandled and unknown-handled inbound attempts. Reorders must match the customer, non-null `PET_ID` and nonblank `PRODUCT_PART_NUMBER`, on a different order placed strictly after its matched aged cancellation through 14 days inclusive. Apply the established non-cancelled-order rules: exclude an entire reorder order if any line has cancellation evidence or unknown status. Reorders may use any approval channel or prescription source. Count each customer once in their segment if any qualifying aged order has a matching reorder in its own follow-up window. Percentages use all customers in the contact segment, including those missing matching keys or without reorders; N/A denotes an empty segment. Contact-day segments include post-cancellation contacts through day $+3$. Observation cutoff: 2026-10-28 00:00 UTC.

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### 2.3.4 Reorders by contact timing

Table 18 compares customers with a qualifying contact on days $-7$ through $+3$ relative to aged cancellation against the rest of the aged-order cohort. The contacting group's 14-day reorder rate is 7.70 percentage points higher.

Table 18: Same-pet, same-product-part-number reorders within 14 days after aged cancellation: customers contacting on days $-7$ through $+3$ versus the rest of the cohort. Original CMPU vet-diet `CUSTOMER_UPLOAD` orders were cancelled May 1–July 31, 2026, inclusive, UTC.

| Customer group | Customers | Customers reordered | Reorder rate (%) |
|---|---|---|---|
| Contacted on days $-7$ through $+3$ | 1,184 | 237 | 20.02% |
| Rest of aged cohort | 29,198 | 3,595 | 12.31% |

*Notes:* A customer belongs to the contacting group if any qualifying inbound contact is linked to one of their original aged orders and the contact customer ID matches the order's customer. Include all contact categories and handled states. Days are elapsed 24-hour bins: the window starts exactly seven days before cancellation and ends just before four days after cancellation, covering all of day $+3$. The complementary group includes customers contacting only outside this window and customers with no qualifying contacts. Each customer belongs to exactly one group; the groups sum to 30,382 customers and 3,832 returning customers. Use the same non-cancelled same-customer, `PET_ID` and `PRODUCT_PART_NUMBER` reorder endpoint as Table 17: a different order placed strictly after its matched aged cancellation through 14 days inclusive. Count a customer once if any qualifying aged order has a matching reorder in its own window; the contact need not concern that same aged order. Percentages use all customers in each group. Observation cutoff: October 28, 2026 at 00:00 UTC. Contacts after cancellation contribute to group membership; this is a descriptive comparison and does not establish a causal effect of contact.

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## 2.4 Aged Order Delay Analysis

An order becomes an aged order when canceled automatically after the time between order creation and time of shipment or any other cancellation exceeds 11 days. A detailed breakdown of cancellation reasons can be found in Figure 1 for the study population.

The population of orders for this study consists of first orders and first order lines for prescriptions as we would like to analyze issues at prescription upload and approval.

For this analysis; 95,073 vet diet prescriptions created between August 15, 2026 and October 15, 2026 were used with `CUSTOMER_UPLOAD` as the approval channel. This corresponds to 87,724 distinct orders and 95,073 order lines. The 24,408 orders described in `OMS` with the cancellation reason `AGED_ORDER` constitute the aged orders leading to 26,443 line items, 26,443 corresponding prescriptions and 20,587 customers.

The difference in counts between first orders and order lines and all orders and order lines is shown in Table tab:first-all-order-cancellations.

Factors that might be contributing to an order aging are: delay in prescription upload by the customer, delay in processing the upload, and delay in approving the prescription after the uploaded prescription document is accepted.

The data for uploaded documents/entities does not cover all the prescriptions, the coverage is about 31% in this study. The number of prescription documents uploaded for 26,443 aged order prescriptions is 782 ( 3%) compared to 33,336 uploads for 95,073 prescriptions overall ( 35%) pointing to a severe non-response by the customers. See Table 19.

Only 6 aged order related prescription were approved out of 52 accepted/782 uploaded (See Table 20 for comparison with the rest of the order line population.) The median time for these prescriptions to become accepted was 10 days after upload, as a result missing the aged order cancellation deadline (Figure 4). For comparison, the distributions of the time difference between upload of the documents and their final states are in Figure 3 for all entities. Customers seem to upload at similar delays after order creation for overall and aged order prescriptions (Figure 5).

From the available entity upload data, it takes less than a minute to approve an accepted prescription document see Figure 6.

We look at the chance of an order becoming an aged order in Section 2.4.1. Here, we build a survival curve. Survival means that an order has yet to encounter a document upload or cancellation. One can see in Figure 2 that after day 3, the change of an order becoming an aged order is about  50% (28.90

Table 19: Study population and uploaded-document counts. Counts are distinct within each population.

| | | | | | Uploaded documents | |
|---|---|---|---|---|---|---|
| Population | Prescriptions | Orders | Order lines | Customers | Total | Accepted |
| All approval channels, matched to first-order lines | 103,516 | 95,533 | 103,516 | 80,649 | 30,735 | 26,254 |
| `CUSTOMER_UPLOAD`, matched to first-order lines | 95,073 | 87,724 | 95,073 | 73,905 | 29,715 | 25,606 |
| `CUSTOMER_UPLOAD` aged-cancellation subset | 26,443 | 24,408 | 26,443 | 20,587 | 782 | 52 |

*Notes:* CMPU vet-diet prescriptions created from August 15, 2026 inclusive through October 15, 2026 exclusive, UTC. The aged-cancellation subset includes qualifying cancellations observed through October 27, 2026, 00:00 UTC. Line associations and document counts were refreshed on October 28, 2026. Uploaded documents count distinct linked `DIET_PRESCRIPTION` upload entities created August 1–October 25, 2026, UTC, with a matched first-order line; entities shared by prescriptions count once per row. Accepted documents have current `STATE_NAME = ACCEPTED`.

Table 20: Prescription decisions for the study populations.

| Population | Prescriptions | Approved | Unknown | Rejected |
|---|---|---|---|---|
| All approval channels, matched to first-order lines | 103,516 | 36,874 | 66,595 | 47 |
| `CUSTOMER_UPLOAD`, matched to first-order lines | 95,073 | 30,504 | 64,569 | 0 |
| `CUSTOMER_UPLOAD` aged-cancellation subset | 26,443 | 6 | 26,437 | 0 |

*Notes:* Distinct prescriptions in the same populations as Table 19. Approved means `PRESCRIPTION_APPROVED = TRUE`, consistent with the prescription registry's `APPROVED_NUMERATOR`. Rejected means current `RX_STATUS = DENIED` without an approval flag. Unknown includes cancelled and pending prescriptions without a recorded approval or rejection outcome. The three categories are mutually exclusive and sum to the prescription count in each row. Decisions use the current source snapshot refreshed October 28, 2026; approval and denial times are not restricted to the aged-cancellation cutoff.

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Table 21: Cancellation reasons, order counts and net sales for first orders and all linked orders.

| | First orders | | | All orders | | |
|---|---|---|---|---|---|---|
| Cancellation reason | Orders | Lines | Net sales | Orders | Lines | Net sales |
| [Aged order outside cutoff] | 372 | 398 | 33,489.70 | 372 | 398 | 33,489.70 |
| [No recorded cancellation reason] | 38,413 | 40,864 | 3,163,597.96 | 60,879 | 69,510 | 5,372,693.08 |
| Aged order | 24,408 | 26,443 | 2,236,727.20 | 24,409 | 26,445 | 2,236,909.55 |
| Customer initiated cancellation | 11,601 | 13,127 | 995,271.62 | 12,255 | 13,933 | 1,064,806.64 |
| Customer request | 4,890 | 5,486 | 428,933.38 | 5,037 | 5,683 | 446,285.60 |
| Duplicate order | 61 | 64 | 5,237.98 | 74 | 80 | 6,499.90 |
| Fraud | 13 | 15 | 1,809.92 | 13 | 15 | 1,809.92 |
| Inventory issue | 3 | 3 | 264.42 | 3 | 3 | 264.42 |
| No response | 541 | 575 | 48,509.38 | 559 | 593 | 50,252.01 |
| Other | 302 | 336 | 25,662.96 | 317 | 354 | 26,850.17 |
| Payment declined | 2,444 | 2,679 | 221,343.75 | 2,541 | 2,786 | 229,682.76 |
| Payment issue | 15 | 17 | 1,503.07 | 16 | 18 | 1,638.91 |
| Pharmacy rejection | 137 | 157 | 13,237.61 | 137 | 157 | 13,237.61 |
| Prescription issue | 4,524 | 4,909 | 412,858.46 | 4,524 | 4,910 | 412,972.44 |
| Reseller | 1 | 1 | 132.99 | 1 | 1 | 132.99 |
| TOTAL | 87,725 | 95,074 | 7,588,580.40 | 111,137 | 124,886 | 9,897,525.70 |

*Notes:* CMPU vet-diet `CUSTOMER_UPLOAD` prescriptions created August 15–October 14, 2026, UTC. All orders includes first and subsequent linked orders, without an order-placement window. Qualifying aged cancellations are observed on or before October 27, 2026, 00:00 UTC; current aged reasons outside that cutoff are shown separately for reconciliation. Other reasons reflect the comparison snapshot. Counts are distinct orders and physical lines within each group. Net sales sum `ORDER_LINE_NET_SALES` once per shipment/line record, for qualifying lines only. Amounts are displayed to two decimals; five-decimal values remain in the data exports. Gray net-sales amounts have a recorded cancellation reason and represent unrealized sales, not realized revenue. TOTAL includes these gray amounts. The validated comparison snapshot contains one additional matched first-order line/order relative to the earlier frozen population in Table 19.

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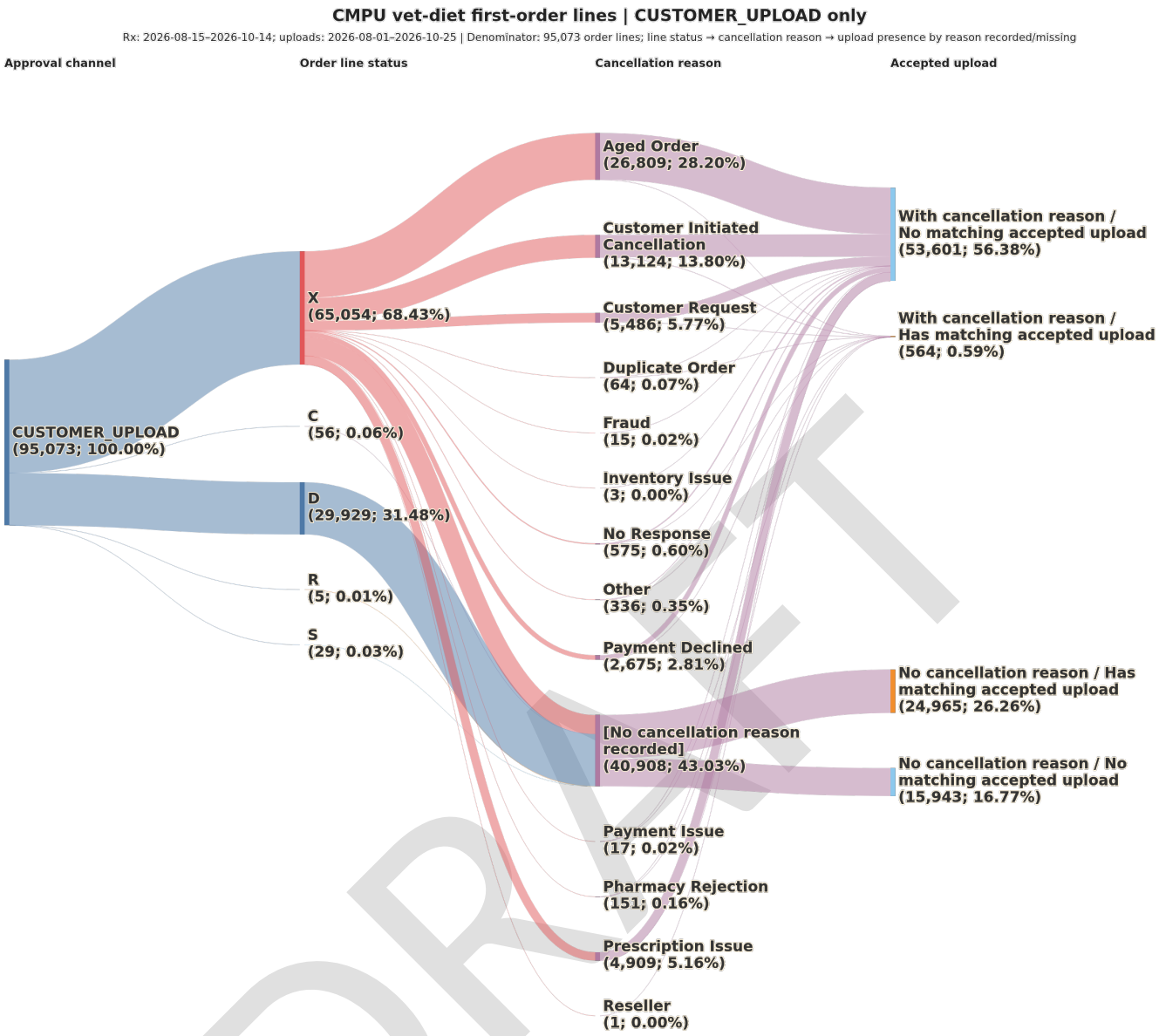



Figure 1: CMPU vet-diet first-order lines: `CUSTOMER_UPLOAD` approval channel.

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### 2.4.1 Survival Analysis

Figure 2 shows the probability that a first-order line is still awaiting an accepted upload or cancellation after order placement.

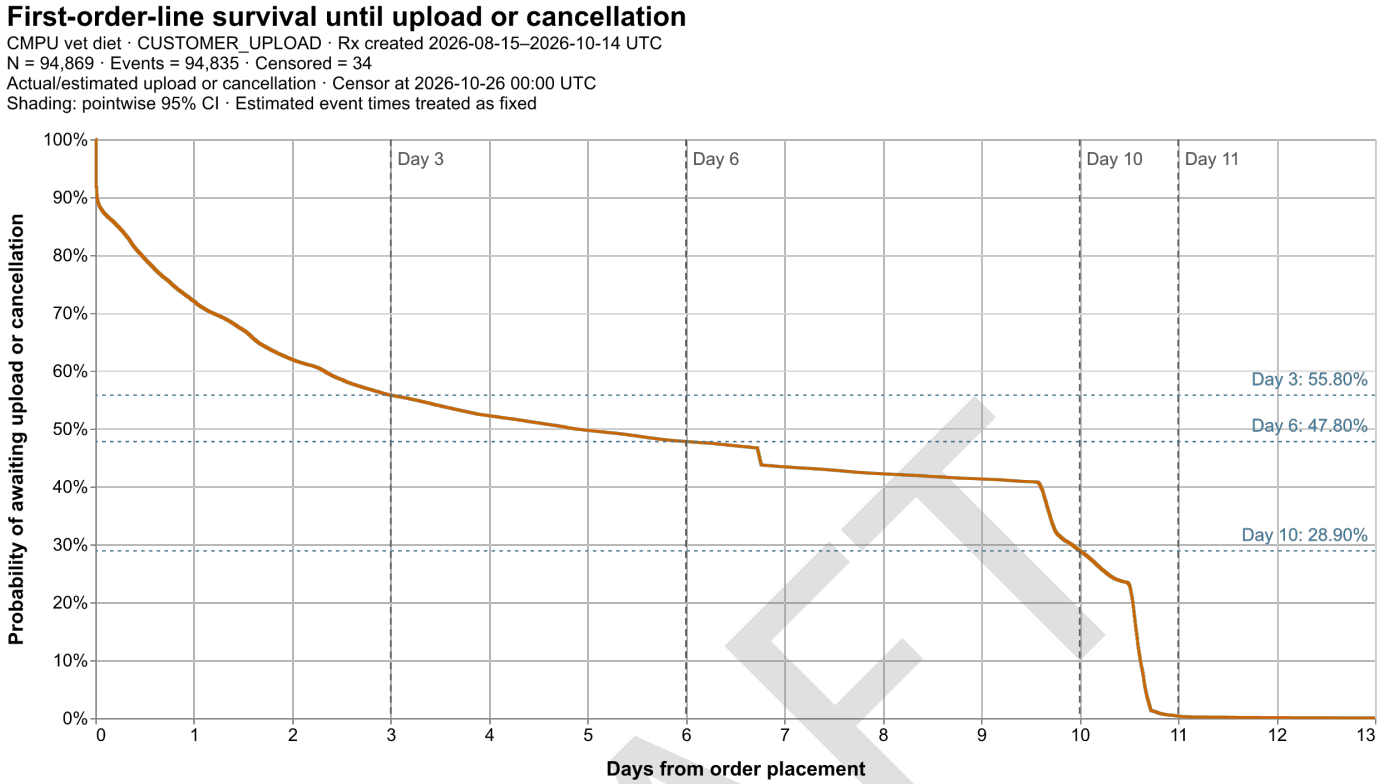



Figure 2: Kaplan–Meier survival curve for 94,869 eligible distinct first-order lines linked to CMPU vet-diet `CUSTOMER_UPLOAD` prescriptions created August 15–October 14, 2026, inclusive, UTC. Time zero is order placement; the event is the earlier of the actual or estimated accepted-entity state-change time and order cancellation. The upload-entity discovery window is August 1–October 25, 2026, inclusive. Missing events and events after October 26, 2026 at 00:00 UTC are right-censored at that cutoff; events exactly at the cutoff are observed. Lines with missing placement, placement after the cutoff or negative event durations are excluded. Shading shows pointwise 95% log-log Greenwood confidence intervals conditional on the supplied event times; it does not include uncertainty from the fixed 0.96 estimation rule. The plot displays days 0–13, with vertical markers at days 3, 6, 10 and 11 and horizontal guides at days 3, 6 and 10. Survival means neither accepted-state change nor cancellation has occurred; it does not measure upload completion alone.

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### 2.4.2 Prescription Document Upload Analysis

Figures 3 and 4 show the elapsed time from upload entity creation to its current recorded state change, grouped by final state, for the overall study-linked population and its aged-order subset. Both figures use the same state order and time scale.

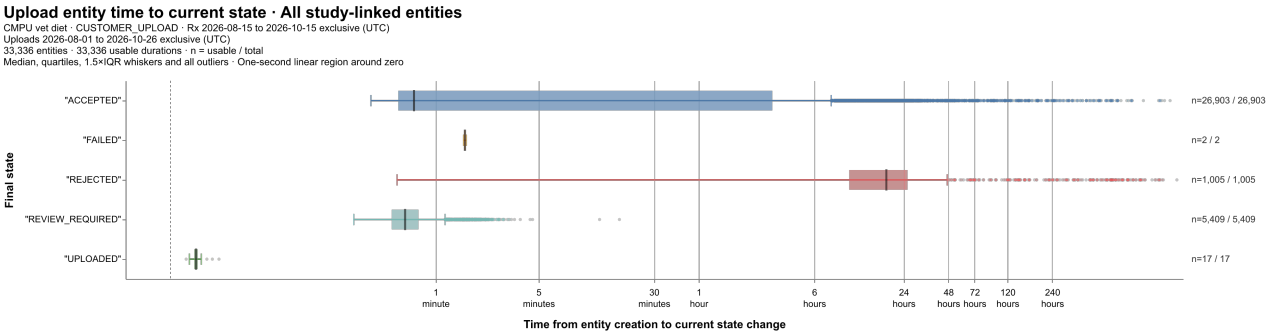



Figure 3: Upload creation to current recorded state change for all study-linked CMPU vet-diet `CUSTOMER_UPLOAD` entities ($N = 33{,}336$). Each upload entity counts once. Boxplots show medians, quartiles, 1.5-times-IQR whiskers and all outliers, grouped by final state.

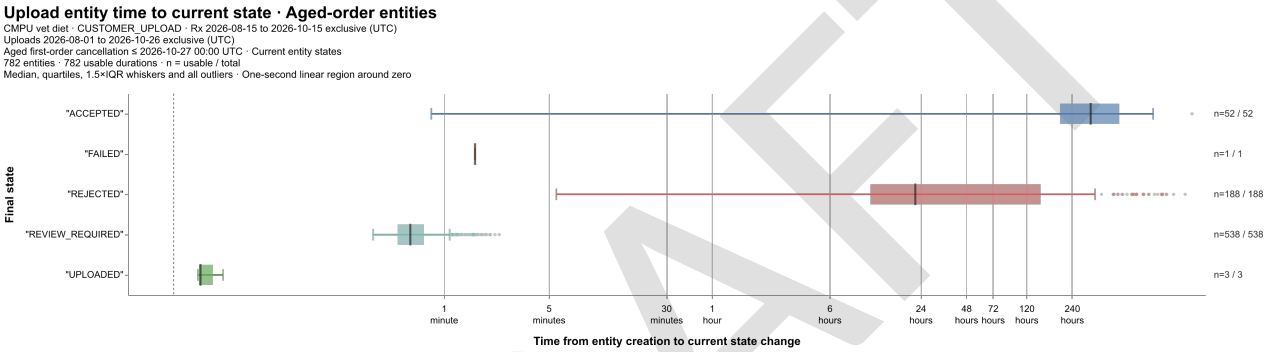



Figure 4: Upload creation to current recorded state change for the aged-order entity subset ($N = 782$). A qualifying matched first-order line was cancelled for aging on or before October 27, 2026, 00:00 UTC. Current entity states and state-change times are not capped at cancellation.

Figure 5 compares upload entity creation relative to order placement for the overall and aged-order populations.

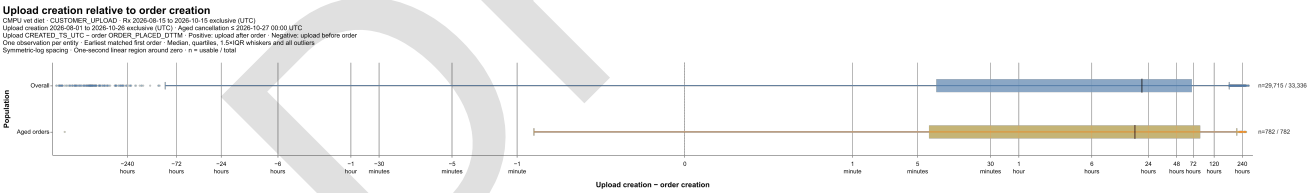



Figure 5: Upload creation relative to order placement for the overall and aged-order entity populations. Elapsed time is upload creation minus order placement: positive values indicate upload after ordering and negative values indicate upload before ordering. Each entity uses its earliest matched first-order placement. Plotted durations include 29,715 overall entities and 782 aged-order entities; entities missing an order-placement timestamp are excluded from the boxplots. Medians, quartiles, 1.5-times-IQR whiskers and all outliers are shown.

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### 2.4.3 Approval Analysis

Figure 6 shows the cumulative distribution of the interval between the selected upload entity's accepted-state change and the latest recorded approval for qualifying first-order lines.

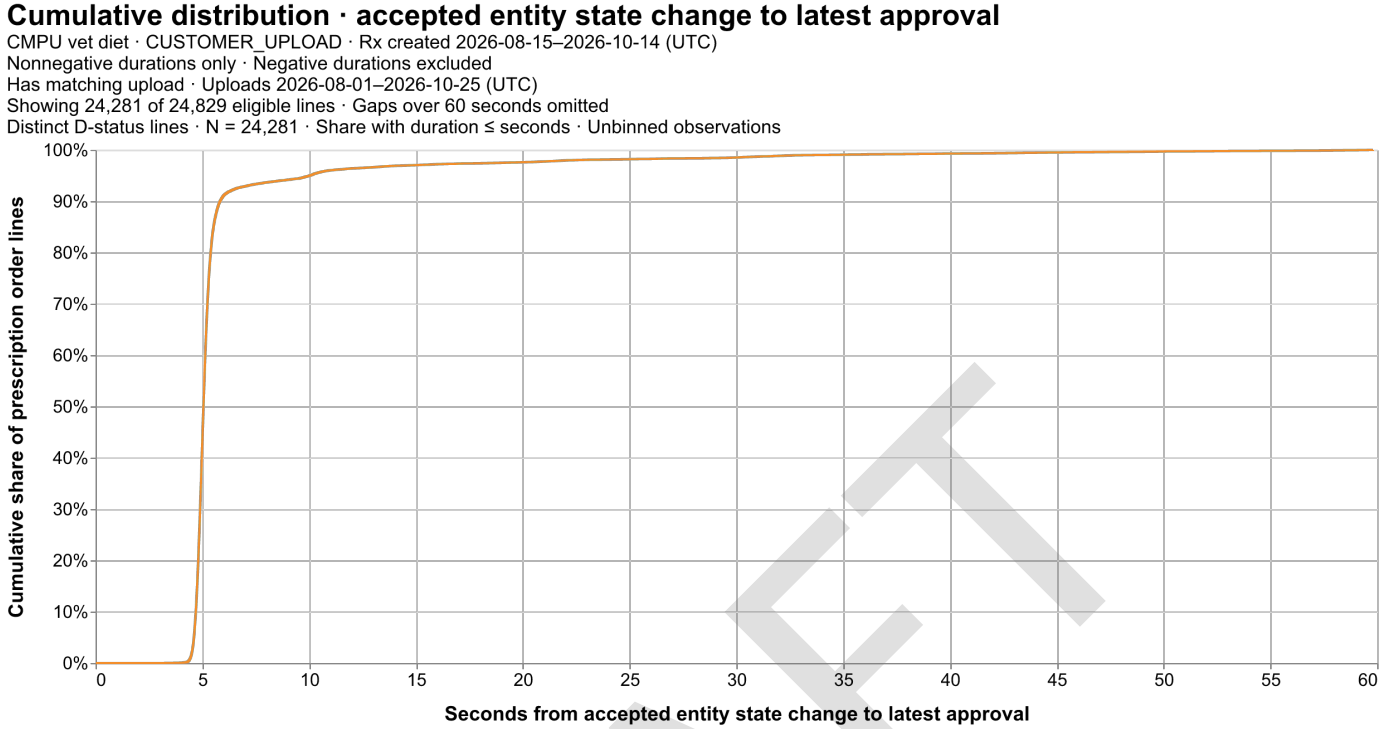



Figure 6: Cumulative distribution of accepted-upload-to-approval time for CMPU vet-diet `CUSTOMER_UPLOAD` first-order lines with current status D. Elapsed time is the latest recorded approval timestamp minus the selected accepted entity's state-change timestamp. The curve shows only nonnegative gaps from 0 through 60 seconds inclusive and is normalized to that subset; gaps above 60 seconds are omitted. Each distinct matched first-order line counts once.

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# 3 References

[1] *RXP CMPU Vet Diet Cancellations Analysis*. GitHub repository. https://github.com/myunt1-chwy/rxp-cmpu-vet-diet-cancellations-analysis.

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